AthenaHealth Interview Experience

Athenahealth is a leading healthcare technology company that provides cloud-based solutions like electronic health records, revenue cycle management, patient engagement, and telehealth services to medical groups and health systems. With offices in Chennai, Pune and Bangalore, the company's vision is to create a thriving ecosystem that delivers accessible, high quality and sustainable healthcare for all.

Role Offered: Product Engineer Intern

Sequence of Selection:

1. Online Test
2. Technical Interview
3. Technical + Managerial Interview
4. HR Interview

1. Online Test:

The online assessment was conducted on the HackerRank platform with a total duration of 80 minutes. The test began with 7 multiple-choice questions (MCQs) on Data Structures and Algorithms (DSA) and Database Management Systems (DBMS). The questions were of easy difficulty and covered topics such as the data structure used for level order traversal, postfix evaluation, and identifying the sorting algorithm for a given array.

Following the MCQs, there were two coding questions of easy-to-medium difficulty:

1. **Minimum Cost after Insertion**
   An array of integers is given, and the cost is defined as:

   $$\text{Cost}=\sum (arr[i] - arr[i+1])^2$$

   The task was to insert one element anywhere in the array such that the overall cost is minimized and return the minimum cost.

   Example:
   Input: [4, 7, 1, 4]
   Initial Cost = $(4-7)^2 + (7-1)^2 + (1-4)^2 = 54$
   By inserting 4 between 7 and 1, the array becomes [4, 7, 4, 1, 4], reducing the cost to 36.

   Hint: For each pair, try inserting the middle element (e.g., for the pair (7, 1), the middle is (7+1)/2 = 4). This can be solved efficiently in a single pass.

2. **Minimum Subarray with K Distinct Integers**
   This problem was similar to LeetCode 992: Subarrays with K Different Integers, but instead of counting all such subarrays, the requirement was to return the length of the minimum subarray that contains exactly *k* distinct integers.

   I was able to complete all the questions within 30 minutes, well ahead of the allotted 80 minutes. After completing the online assessment in the morning, the results were announced by 1 PM, and the interviews were scheduled at 2 PM on the same day. A total of 31 students were shortlisted for the interview round.

2. Technical Interview:

The first round began with a self-introduction, followed by a discussion on my projects. I explained one of my projects in detail, and the interviewer asked several follow-up questions based on it. My suggestion for future candidates would be to choose a project you are confident about and know it thoroughly.

Next, I was asked a few fundamental **DBMS concepts**, including:

- Explanation of **ACID properties**
- Definition of **Normalization** and details of **1NF, 2NF, and 3NF**
- Difference between **SQL and NoSQL databases**, along with which I had used in my projects and the reasons behind my choice

The interviewer also tested my understanding of **Operating Systems concepts**, such as processes and threads. This was followed by a **SQL problem**:

Given two tables, find the employees whose total order amount is greater than 10,000. I was able to write the query successfully.

Later, I was asked to solve and explain two coding problems:

1. Implement Stack using Queues
2. Remove Nth Node From End of List

I explained both the brute-force and optimized solutions and was able to write the code, which satisfied the interviewer. The round concluded with me asking questions about Athenahealth's work culture and the role of AI within the organization. The interviewer was friendly, and the interaction went smoothly.

After around 30 minutes, I was asked to join the second round.

3. Technical + Managerial Interview:

The second round began with a self-introduction and a brief discussion about my family members. Although this was expected to be a managerial round, I was given two coding problems instead.

1. **Find the Missing Elements in an Array**
    - Initially, I explained the brute-force approach using a set to identify the missing element.
    - I then provided the optimized approach by applying the formula for the sum of the first *n* natural numbers and subtracting the actual sum of the array elements.
    - As a follow-up, the interviewer asked how I would handle the case where two numbers are missing. I suggested marking the indices of present elements with negative values and then traversing again to identify the positive indices. However, during the dry run, we identified that my approach failed in certain edge cases. I attempted to resolve it but could not arrive at the complete solution within the time.
2. **Nuts and Bolts Problem**
    - The task was to match nuts and bolts of the same size.

- I first explained the brute-force approach of comparing each nut with every bolt, resulting in **$O(n^2)$** complexity.
- I then optimized it to **$O(n)$** using a hashmap, which satisfied the interviewer.

The round concluded with me asking the interviewer multiple questions about the role and the company. My advice to future candidates is to be honest—if you cannot come up with the optimal solution immediately, explain your thought process and mention that you would need more time to refine it.

At the end of this round, 12 candidates were shortlisted for the HR round, which was scheduled for the next morning

3. HR Interview:

This round primarily assessed how well I aligned with the role and the company culture. It began with a self-introduction and a brief discussion about my family members. The interviewer then asked me to explain one of my projects in depth, followed by questions on why I chose Athenahealth, why I would be a good fit for the role, and a few follow-ups based on my resume. The session lasted around 20 minutes and was smooth, as the interviewer was very approachable and friendly.

Later, the results were announced, and six of us received offers.

For future candidates, I would suggest being well-prepared with your self-introduction, internship experiences (if any), and project details. Communicate with clarity and confidence. Even if you don't know the answer to a question, stay calm, as interviewers often guide you toward the right direction. Consistent preparation and strong core concepts are key. This was my fifth interview, and my continuous efforts helped me succeed here. Don't be discouraged if you're not selected—the right opportunity will come your way. Wishing you all the best!

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